# LINEAR PROGAMMING: SOME EXERCISE PROBLEMS

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Convert the following problem to an LP:

Minimize $|x| + |y| + 3|z|$

subject to

$$x + 2y \leq 10$$
$$2x + 5z = 60$$

**Solution**

Let us introduce variables a, b and c, which are $\geq 0$.

Suppose

$$|x| \leq a$$
$$|y| \leq b$$
$$|z| \leq c$$

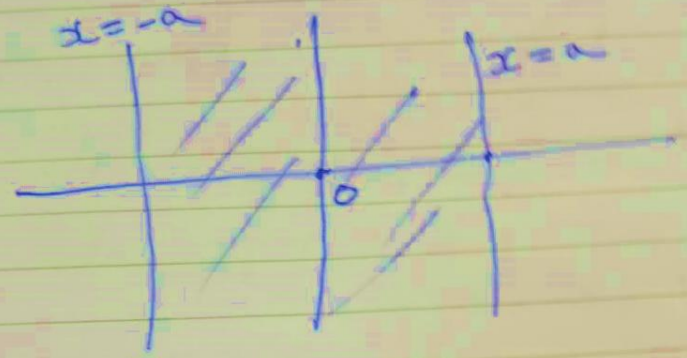
then minimize $a + b + 3c$ can be made objective functia.

$$|x| + |y| + 3|z| \leq a + b + 3c$$

But we have to add constraints

$|x| \leq a \Rightarrow x \leq a$ and $x \geq -a$

this is to say that $x$ lies between $[-a, a]$.




Same thing can be done for $y$ and $z$.

$$x \geq -a$$

$$\Rightarrow x - s = -a$$

where $s$ is a unphysical variable. $s \geq 0$

Or $$s - x = a$$

$x$ Unrestricted means $x = x^+ - x^-$ where $x^+, x^- \geq 0$

Q.

$f(x) = \max(3x+4y, 6x+y)$

Minimize $f(x)$

Subject to

$5x + 2y \leq 7$

$7x + 9y \geq 8$

$x, y \geq 0$

Solution

Define $\theta$ such that

$\theta \geq 3x+4y$

$\theta \geq 6x+y$

where $\theta \geq 0$.

The above are two linear constraints.

Change the objective fn to minimize $\theta$.

A free variable $x$ can be replaced by $x^+ - x^-$ where $x^+ \geq 0$, $x^- \geq 0$

Another way be eliminat it. Like

$$x + y = 5$$

$$\Rightarrow x = 5 - y$$

Minimize $3x + 2y + 4z$

Subject to

$$x + y + z = 10$$

$$x + 2y = 7$$

$$0 \leq x \leq 5$$

$$0 \leq y \leq 8$$

$$0 \leq z \leq 10$$

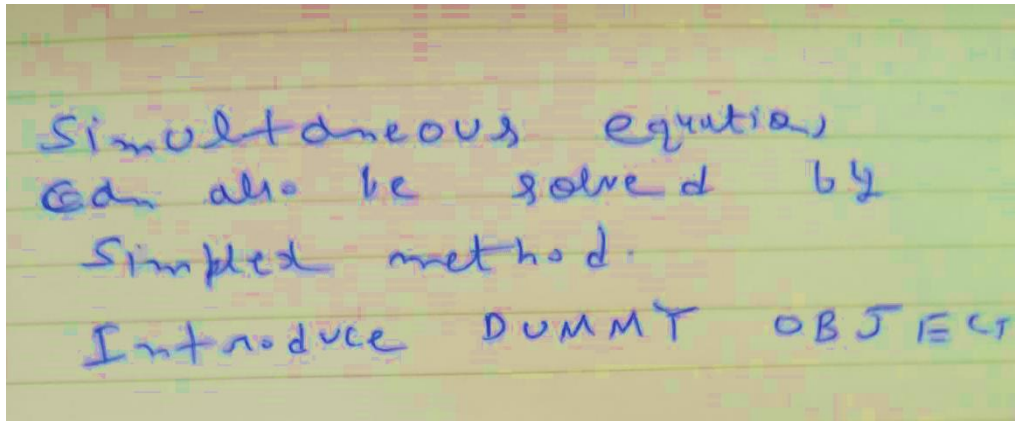
Write the last three constraints as

$$x + s_1 = 5$$

$$y + s_2 = 8$$

$$z + s_3 = 10$$

$$x, y, z, s_1, s_2, s_3 \geq 0$$


Simultaneous equation
Can also be solved by
Simplex method.
Introduce DUMMY OBJECT

Q.: Choose the correct option. In a minimization problem, the optimum function value came out to be 100. If one of the design variables is constrained to be an integer, the optimum value will be

(A) more than 100 (B) less than 100 (C) less or equal to 100 (D) more or equal to 100.

(D) more or equal to 100

Why? I am constraining it to be integer, which is extra constraint. Hence, solution will deteriorate. I was getting 100 as minimum, now I may not be able to get 100. Surely I shall not get 100 because that is a better solution of minimization.

A function $f(\boldsymbol{x})$ is said to be strictly convex if for any two distinct points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$

$$f\{\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2\} < \lambda f\{\boldsymbol{x}_1\} + (1-\lambda) f\{\boldsymbol{x}_2\}.$$

Between which values $\lambda$ lies?

# 0 and 1

Consider the problem of maximization of a function with a constraint $g(\boldsymbol{x}) \geq b$. If $b$ is increased, $f$ is likely to increase or decrease?

Objective function is likely to decrease because you are making constraint stricter (more tight). Hence, you cannot expect a better solution.

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

s.t.

$$x_1 + 2x_2 + 3x_3 = 15$$
$$2x_1 + x_2 + 5x_3 = 20$$
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Solve by big M method.

Solution $x_1 = \frac{15}{6}$, $x_2 = \frac{15}{6}$

$x_3 = \frac{15}{6}$, $x_4 = 0$

$$Z = \frac{15}{06} + 2 \times \frac{15}{06} + 3 \times \frac{15}{06} - 0 = 15$$

First constraint

$x_1 + 2x_2 + 3x_3$

$= \frac{15}{6} + 2 \times \frac{15}{6} + 3 \times \frac{15}{6} = 15 = RHS$

Hence, constraint is active.

Similarly we can see that other constraints are also active.

In fact, all equality constraints have to be active and they may have non-zero shadow price.

Maximize $Z = x_1 + 2x_2$

s.t.

$$3x_1 + 2x_2 \leq 12$$

$$x_1 = 2$$

$$x_1 + x_2 \geq 8$$

$$-x_1 + x_2 \geq 4$$

$$x_1, x_2 \geq 0$$

This problem is infeasible.

Maximize $Z = 15x_1 + 25x_2$

s.t.

$$4x_1 + 6x_2 \leq 360$$

$$x_1 \leq 60$$

$$x_2 \leq 40$$

$$x_1, x_2 \geq 0$$

Solution is $x_1 = 60, x_2 = 0$.

What is the unit worth of resource corresponding to $4x_1 + 6x_2 \leq 360$

L.H.S. $= 4x_1 + 6x_2 = 4 \times 60 + 6 \times 0$

$= 240 \neq 360$ (RHS)

Hence, the constraint is inactive.

Shadow price $= 0$